# Thick Cloud Removal for Multitemporal Remote Sensing Images: When Tensor Ring Decomposition Meets Gradient Domain Fidelity

Li-Yuan Li, Ting-Zhu Huang, Yu-Bang Zheng, Wen-Jie Zheng, Jie Lin, *Graduate Student Member, IEEE*, Guo-Cheng Wu, and Xi-Le Zhao

***Abstract*— Thick clouds in remote sensing (RS) images deteriorate the visual quality and hinder subsequent applications. The emerging multitemporal RS images with rich temporal information bring the opportunity for cloud removal. How to effectively exploit the rich temporal information of the multitemporal RS images remains challenging. As multitemporal RS images with the same geographic scene, the spatial gradient of RS images at different time nodes has a resemblance, which can guide the reconstruction of the cloudy region. Motivated by this, we suggest a gradient domain fidelity with respect to the guided gradient for thick cloud removal in multitemporal RS images, which faithfully preserves the fine edges and textures compared with the original pixel domain fidelity. Armed with the gradient domain fidelity, we propose a low-rank tensor ring decomposition model (termed as TRGFid) for the thick cloud removal problem. In the proposed model, the guided gradient of the cloudy region is availably estimated by using the Regression method from the cloud-free region of different time nodes. Moreover, we develop an efficient proximal alternating minimization (PAM)-based algorithm for solving the proposed nonconvex model. Extensive simulated and real experiments show that the proposed method outperforms its competitors and preserves fine edges and textures.**

***Index Terms*— Cloud removal, gradient domain fidelity, multitemporal remote sensing (RS) images, proximal alternating minimization (PAM), tensor ring (TR) decomposition.**

## I. Introduction

With the rapid development of remote sensing (RS) technology and applications, RS images acquired by optical RS detection contain rich spatial, spectral, and temporal information, making them widely used in many applications, such as classification [1], [2], [3], unmixing [4], [5], [6], fusion [7], [8], [9], and target detection [10], [11], [12]. Unfortunately, during the acquisition or transmission process, thick clouds/shadows in real-world RS images deteriorate the visual quality, severely impacting subsequent applications. Therefore, developing effective cloud removal methods is necessary to faithfully reconstruct ground-truth information from cloud-contaminated RS images.

Researchers have proposed many methods for cloud removal of RS images [13], [14], [15], [16]. The existing methods include blind methods and nonblind methods. Blind methods, such as [17], [18], and [19], aim to reconstruct the cloudy region without known cloud/shadow masks. Furthermore, with the rocketing development of cloud detection techniques for RS images, a large amount of cloud/shadow masks are available. Nonblind methods utilize the given cloud/shadow masks, which can be classified into spatial-based, spectral-based, temporal-based, and hybrid-based methods.

The spatial-based methods utilize image inpainting techniques [20], [21], which only rely on the image prior in the spatial dimension. The methods assume similar texture features in cloud-contaminated and cloud-free regions and reconstruct missing information by local or nonlocal pixels of cloud-free regions. The representative methods include interpolation methods [22], [23], partial differential equation methods [24], [25], variational methods [26], [27], and exemplar-based methods [28], [29]. However, these methods obtain inaccuracy reconstructed results when the missing area of RS images is large or contaminated by thick clouds, since there is not enough reference information. The spectral-based methods utilize the image prior in the spectral dimension. For example, the methods in [30], [31], [32], [33], and [34] used data from other relevant bands to reconstruct the cloudy regions of Aqua Moderate Resolution Imaging Spectroradiometer at the sixth band. However, in the thick cloud removal task, the absence of the same position information on all bands leads to the inability of these methods to deal with this task. Considering satellites can obtain the same regions of RS images at different time nodes, while clouds cover the images at different locations at different time nodes. The temporal-based methods aim to recover the cloudy regions using the correlation between each temporal image of multitemporal images. Zeng et al. [35] used the linear relationship between multitemporal images

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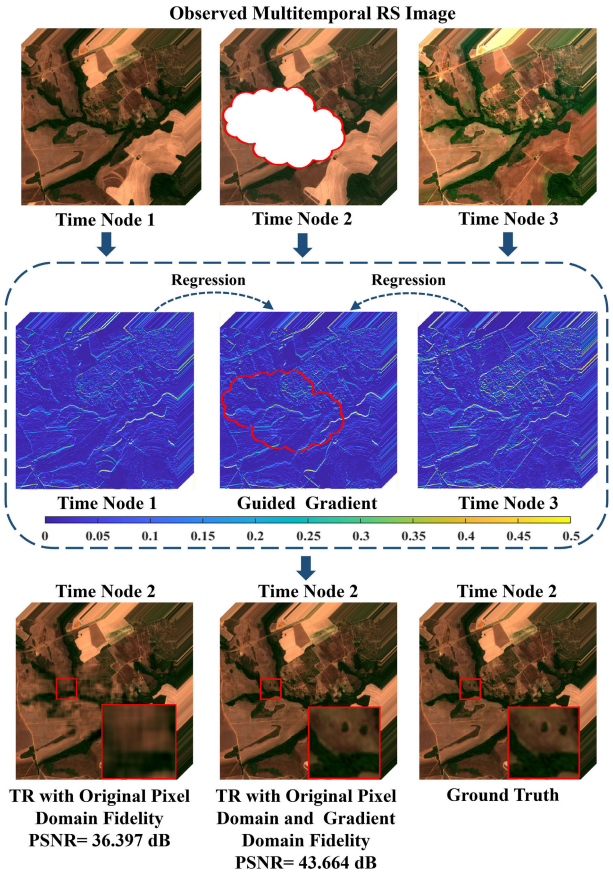



Fig. 1. Illustration of the proposed gradient domain fidelity. The first row is the observed RS images at different time nodes, the second row is the spatial gradient along the horizontal direction of the images at time nodes 1 and 3 and the guided gradient along the horizontal direction of the images at time node 2, and the third row is the recovered images and ground truth at time node 2.

and corresponding missing pixels and proposed a weighted linear Regression model to recover the missing pixels. Wang et al. [36] exploited temporal smoothness and low rankness and suggested a temporally contiguous robust matrix completion model. The aforementioned three classes of methods utilize the image prior in only one dimension but ignore the prior in the other dimensions.

To fully exploit the image prior, hybrid methods using image priors in two of the spatial, spectral, and temporal dimensions are proposed. For joint spatial–temporal methods, Chen et al. [37] developed a spatially and temporally weighted Regression-based method to recover cloud-free Landsat images. For joint spectral–temporal methods, Melgani [38] proposed a contextual prediction process to reconstruct each missing measurement area by exploiting the local spectral–temporal relationships. For joint spatial–spectral methods, Benabdelkader and Melgani [39] presented a post-reconstruction method, which captures spatial–spectral correlations to improve the contextual reconstruction process. To use all the available image priors, hybrid methods using image priors in spatial–spectral–temporal dimensions are proposed. In the past decades, by using low-rank tensor decomposition [40], such as CANDECOMP/PARAFAC (CP) decomposition [41], Tucker decomposition [42], and tensor singular value decomposition (t-SVD) [43], many low-rank tensor completion methods [44], [45], [46], [47], [48], [49], [50], [51] have been proposed to remove clouds of multitemporal RS images effectively. Yokota et al. [52] applied the smoothness constraints on the CP factors of CP decomposition and proposed the corresponding tensor completion model. Liu et al. [53] introduced a sum of the nuclear norm tensor completion method based on Tucker decomposition. Ji et al. [54] proposed a nonlocal Tucker decomposition method, which can capture the nonlocal self-similarity of RS images. Chen et al. [55] used the tensor nuclear norm (TNN) to capture the global low rankness and introduced a spatial–spectral continuity constraint in the proposed model. Zhang et al. [56] proposed a thick cloud removal method of multitemporal RS images based on t-SVD and integrated a deep spatiotemporal prior. Recently, tensor network decomposition, including tensor train (TT) decomposition [57] and tensor ring (TR) decomposition [58], obtained satisfactory results in the tensor completion problem. Bengua et al. [59] proposed a TT decomposition-based tensor completion method. TR decomposition can fully characterize the correlation of spatial–spectral–temporal in multitemporal RS images [58]. Yuan et al. [60] introduced nuclear norm regularization on the latent TR factors, and He et al. [61] combined a spatial local continuity constraint with TR decomposition. However, the previous methods only consider the pixel domain fidelity, which usually leads to details blurriness; see Fig. 1 for a demo.

To tackle this issue, we suggest a gradient domain fidelity as a complement to the original pixel domain fidelity, which makes full use of the gradient[1] information of multitemporal RS images at cloud-free time nodes. The motivation is that the gradient of multitemporal RS images at different time nodes has a resemblance since the multitemporal RS images with the same geographic scene. Thus, the gradient of the cloudy region is availably estimated from the cloud-free region of different time nodes, which guides the reconstruction of the cloudy region. Specifically, this article makes three contributions as follows.

1) Based on the above observation, we suggest a gradient domain fidelity with respect to the guided gradient as a complement to the original pixel domain fidelity for removing thick clouds of multitemporal RS images, which faithfully preserves the edges and textures compared with the original pixel domain fidelity.
2) Armed with the gradient domain fidelity, we propose a low-rank TR decomposition model (TRGFid) for removing thick clouds of multitemporal RS images. Low-rank TR decomposition can capture the overall structure of the image, while gradient domain fidelity can preserve fine details of the image, especially the edges and textures. The gradient domain fidelity and the low-rank TR regularization are organically integrated to deliver promising results, as shown in Fig. 1.
3) We develop an efficient proximal alternating minimization (PAM)-based algorithm for solving the proposed

[1]Throughout this article, the term "gradient" refers specifically to "spatial gradient."

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TRGFid model. Extensive simulated and real experiments show that the proposed TRGFid outperforms its competitors, especially in edge and texture preservation.

We briefly describe the framework for the remainder of this article as follows. In Section II, we give some notations and preliminary presentations. In Section III, we propose the TRGFid model with the PAM algorithm for removing thick clouds of multitemporal RS images. In Section IV, we present the experimental results and discuss the details. In Section V, we summarize this article.

## II. Preliminaries

### A. Notations

In this section, we summarize some notations used in our paper in the Nomenclature. In addition, we introduce the definitions of mode-$k$ unfolding and $k$-mode product.

*Definition 1 (Mode-k Unfolding [62]):* For an $N$th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, its mode-$k$ unfolding is denoted by $\mathbf{X}_{(k)} \in \mathbb{R}^{I_k \times \prod_{i \neq k} I_i}$, where $\mathcal{X}(i_1, i_2, \ldots, i_N)$ maps to $\mathbf{X}_{(k)}(i_k, j)$ satisfying $j = 1 + \sum_{s=1, s\neq k}^{N} (i_s - 1) J_s$ with $J_s = \prod_{m=1, m\neq k}^{s-1} I_m$. We use $\mathbf{X}_{(k)} = \text{Unfold}_k(\mathcal{X})$ and $\mathcal{X} = \text{Fold}_k\left(\mathbf{X}_{(k)}\right)$ to denote the mode-$k$ unfolding and inverse operator, respectively.

*Definition 2 (k-Mode Product [62]):* The $k$-mode product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and a matrix $\mathbf{A} \in \mathbb{R}^{J \times I_k}$ is defined as $\mathcal{Y} = \mathcal{X} \times_k \mathbf{A} \in \mathbb{R}^{I_1 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_N}$, whose entries are denoted by

$$\mathcal{Y}(i_1, \ldots, i_{k-1}, j, i_{k+1}, \ldots, i_N) = \sum_{i_k=1}^{I_k} \mathcal{X}(i_1, i_2, \ldots, i_N) \mathbf{A}(j, i_k).$$

Based on Definitions 1 and 2, we have

$$\mathcal{Y} = \mathcal{X} \times_k \mathbf{A} \Leftrightarrow \mathbf{Y}_{(k)} = \mathbf{A}\mathbf{X}_{(k)}.$$

### B. TR Decomposition

For a fourth-order tensor $\mathcal{X} \in \mathbb{R}^{M \times N \times B \times T}$, the TR decomposition decomposes $\mathcal{X}$ into four third-order factor tensors, i.e., $\mathcal{G}^{(1)} \in \mathbb{R}^{R_1 \times M \times R_2}$, $\mathcal{G}^{(2)} \in \mathbb{R}^{R_2 \times N \times R_3}$, $\mathcal{G}^{(3)} \in \mathbb{R}^{R_3 \times B \times R_4}$, and $\mathcal{G}^{(4)} \in \mathbb{R}^{R_4 \times T \times R_1}$. Here, $\mathbf{r} = (R_1, R_2, R_3, R_4)$ is called TR rank. The elementwise form of $\mathcal{X}$ can be represented by

$$\begin{aligned}\mathcal{X}(m, n, b, t) = &\sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \sum_{r_3=1}^{R_3} \sum_{r_4=1}^{R_4} \\ &\left\{\mathcal{G}^{(1)}(r_1, m, r_2) \mathcal{G}^{(2)}(r_2, n, r_3)\right. \\ &\left.\mathcal{G}^{(3)}(r_3, b, r_4) \mathcal{G}^{(4)}(r_4, t, r_1)\right\}.\end{aligned}$$

In this article, the TR decomposition is simply denoted by $\mathcal{X} = \Phi(\mathcal{G})$, where $\mathcal{G} = \left\{\mathcal{G}^{(1)}, \mathcal{G}^{(2)}, \mathcal{G}^{(3)}, \mathcal{G}^{(4)}\right\}$. More details of the TR decomposition are given in [58].

## III. Proposed Model and Solving Algorithm

A multitemporal RS image can be expressed as a fourth-order tensor $\mathcal{X} \in \mathbb{R}^{M \times N \times B \times T}$, where $M$ is the spatial height, $N$ is the spatial width, $B$ is the number of spectral bands, and $T$ is the number of time nodes. Establishing appropriate regularization for capturing the prior of multitemporal RS images is the key to the ill-posed problem of cloud removal. Since multitemporal RS images contain plenty of highly similar and repeated information, the TR decomposition-based method [58] was proposed to characterize the spatial–spectral–temporal correlation. However, it only considers fidelity in the pixels domain and cannot preserve the edge features well, which usually leads to details blurriness. To tackle this issue, we suggest a gradient domain fidelity and propose a new method by arming the suggested gradient domain fidelity with low-rank TR decomposition for removing thick clouds of multitemporal RS images.

TABLE I
Notation Declarations

| Notations | Interpretations |
|---|---|
| $x$, $\mathbf{x}$, $\mathbf{X}$, $\mathcal{X}$ | Scalar, vector, matrix, tensor. |
| $\mathcal{X}(i_1, i_2, \cdots, i_n)$ | The $(i_1, i_2, \cdots, i_n)$th element of an $N$th-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$. |
| $\|\mathcal{X}\|_F$ | $\|\mathcal{X}\|_F = \sqrt{\sum_{i_1, i_2, \cdots, i_n} \|\mathcal{X}(i_1, i_2, \cdots, i_n)\|^2}$. |
| $\mathcal{X} \in \mathbb{R}^{M \times N \times B \times T}$ | A multitemporal RS image, where $M$ is spatial height, $N$ is spatial width, $B$ is the number of spectral bands, and $T$ is the number of time nodes. |
| $\mathcal{X}^t$ | RS image at the $t$th time node of $\mathcal{X} \in \mathbb{R}^{M \times N \times B \times T}$. |
| $\mathbf{X}_b^t$ | RS image at the $t$th time node and the $b$th band of $\mathcal{X} \in \mathbb{R}^{M \times N \times B \times T}$. |
| $\Omega$ | Index set of the position of cloud-free pixels. |
| $\Omega^c$ | Index set of the position of cloud-contaminated pixels. |
| $\mathcal{P}_\Omega(\cdot)$ | The projection operator mapping the pixels at $\Omega$ to themselves and setting the other pixels to 0. |
| $\odot$ | Component-wise multiplication. |
| $\oslash$ | Component-wise division. |
| $\otimes$ | Kronecker product. |

### A. Gradient Domain Fidelity

In this section, we suggest a gradient domain fidelity as a complement to the original pixel domain fidelity. In the multitemporal RS images, since the images at different time nodes reflect the same geographic scene, the spatial gradient of RS images at different time nodes has a resemblance. Therefore, the gradient can guide the reconstruction of the cloudy regions. With the guided gradient $\mathcal{Q}_k$ for $k = 1$ and 2 estimated by the observed multitemporal RS image, the gradient domain fidelity can be formulated as follows:

$$\sum_{k=1}^{2} \|\mathcal{X} \times_k \mathbf{D}_k - \mathcal{Q}_k\|_F^2 \tag{1}$$

where $\mathcal{X}$ is the reconstructed image and $\mathbf{D}_k$ is the first-order difference matrix.

Now, let us discuss how to estimate the guided gradient $\mathcal{Q}_k$, which is an important ingredient of the gradient domain fidelity. The concrete procedure consists of four steps.

First, assuming that the observed multitemporal RS image $\mathcal{Y} \in \mathbb{R}^{M \times N \times B \times T}$, where $\mathcal{Y}^{i_1}, \mathcal{Y}^{i_2}, \ldots, \mathcal{Y}^{i_n}$ are images at cloud-free time nodes and $\mathcal{Y}^{j_1}, \mathcal{Y}^{j_2}, \ldots, \mathcal{Y}^{j_m}$ are images at cloud-contaminated time nodes ($n + m = T$), we calculate the

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**Algorithm 1** Algorithm for Estimating the Guided Gradient

**Input:** The observed multitemporal RS image $\mathcal{Y}$, and the index $\Omega$.

**for** $\mathcal{Y}^{i_1}, \mathcal{Y}^{i_2}, \cdots, \mathcal{Y}^{i_n}$ **do**
  Compute $\mathcal{Y}^{\bar{f}}$ via (2).
**end for**
**for** $t = j_1, j_2, \cdots, j_m$ **do**
  **for** $b = 1 : B$ **do**
    Estimate $\hat{Y}_b^t$ via (3).
  **end for**
**end for**
Compute guided image $\hat{X}$.
Compute guided gradient $\mathcal{Q}_k$ for $k = 1, 2$.

**Output:** The guided gradient $\mathcal{Q}_k$.

**Algorithm 2** PAM-Based Algorithm for Solving (4)

**Input:** The observed multitemporal RS image $\mathcal{Y}$, the index $\Omega$, the rank R, regularization parameters $\alpha$, and $\rho = 0.1$.

**Initialization:** $l = 0$, $l_{\max} = 300$, and $\mathcal{X}^{(0)} = \mathcal{Y}$.

Estimate the guided gradient $\mathcal{Q}_k$ for $k = 1, 2$ by Algorithm 1.
**while** not converged and $l < l_{\max}$ **do**
  Update $\mathcal{G}^{l+1}$ via (8).
  Update $\mathcal{X}^{l+1}$ via (14).
  Let $l = l + 1$.
  Check the convergence condition: $\|\mathcal{X}^{l+1} - \mathcal{X}^l\|_F^2 / \|\mathcal{X}^l\|_F^2 \le 10^{-4}$.
**end while**

**Output:** The reconstructed image $\mathcal{X}$.

average image $\mathcal{Y}^{\bar{f}}$ from the images at cloud-free time nodes by

$$\mathcal{Y}^{\bar{f}} = \frac{1}{n}\sum_{k=1}^{n} \mathcal{Y}^{i_k}. \tag{2}$$

Second, we use $\mathcal{Y}^{\bar{f}}$ to estimate the images at cloud-contaminated time nodes $\hat{\mathcal{Y}}^{j_1}, \hat{\mathcal{Y}}^{j_2}, \ldots, \hat{\mathcal{Y}}^{j_m}$ band by band. That is,

$$\hat{Y}_b^t = \mathbf{W}_b^t \mathbf{Y}^{\bar{f}}{}_b + \mathbf{c}_b^t \quad \text{for} \quad t = j_1, j_2, \ldots, j_m \tag{3}$$

where $\mathbf{W}_b^t$ and $\mathbf{c}_b^t$ are the coefficients. The image $\mathbf{Y}_b^t$ contains cloud-free regions and cloudy regions, and thus, we use the least-squares Regression on the cloud-free regions to estimate $\mathbf{W}_b^t$ and $\mathbf{c}_b^t$. Then, the pixel values on the cloudy regions can be estimated by (3).

Third, we replace the cloudy regions in the observed multitemporal RS image with the corresponding regions of $\hat{\mathcal{Y}}^{j_1}, \hat{\mathcal{Y}}^{j_2}, \ldots, \hat{\mathcal{Y}}^{j_m}$ to obtain the guided multitemporal RS image $\hat{\mathcal{X}}$.

Finally, we obtain the guided gradient $\mathcal{Q}_k$ for $k = 1$ and 2 by $\mathcal{Q}_k = \hat{\mathcal{X}} \times_k \mathbf{D}_k$.

The process for estimating the guided gradient is summarized in Algorithm 1.

### *B. TRGFid-Based Cloud Removal Model and Algorithm*

Armed with the gradient domain fidelity, we propose a low-rank TR decomposition model for removing thick clouds of multitemporal RS images, and the model is formulated as follows:

$$\begin{aligned} &\min_{\mathcal{G},\mathcal{X}} \ \frac{1}{2}\|\mathcal{X} - \Phi(\mathcal{G})\|_F^2 + \frac{\alpha}{2}\sum_{k=1}^{2}\|\mathcal{X} \times_k \mathbf{D}_k - \mathcal{Q}_k\|_F^2 \\ &\text{s.t.} \ \ \mathcal{P}_\Omega(\mathcal{X} - \mathcal{Y}) = 0 \end{aligned} \tag{4}$$

where $\alpha$ is the regularization parameter.

Since problem (4) is nonconvex, we introduce the PAM algorithm [63] to solve it. According to the PAM framework, the solution to problem (4) can be obtained by alternately updating

$$\begin{cases} \mathcal{G}^{(n),l+1} = \arg\min\limits_{\mathcal{G}^{(n)}} \left\{ f(\mathcal{G}^{(n)}, \mathcal{X}^l) + \dfrac{\rho}{2}\|\mathcal{G}^{(n)} - \mathcal{G}^{(n),l}\|_F^2 \right\} \\ \mathcal{X}^{l+1} = \arg\min\limits_{\mathcal{X}} \left\{ f(\mathcal{G}^{l+1}, \mathcal{X}) + \dfrac{\rho}{2}\|\mathcal{X} - \mathcal{X}^l\|_F^2 \right\} \end{cases} \tag{5}$$

where $f(\mathcal{G}, \mathcal{X})$ is the objective function of (4) and $\rho > 0$ is the proximal parameter.

*1) Update $\mathcal{G}^{(n)}$:* The $\mathcal{G}^{(n)}$ subproblem is

$$\begin{aligned} \min_{\mathcal{G}^{(n)}} \ \frac{1}{2}\left\|\mathcal{X} - \Phi\left(\mathcal{G}^{(1:n-1),l+1}, \mathcal{G}^{(n)}, \mathcal{G}^{(n+1:N+1),l}\right)\right\|_F^2 \\ + \frac{\rho}{2}\left\|\mathcal{G}^{(n)} - \mathcal{G}^{(n),l}\right\|_F^2. \end{aligned} \tag{6}$$

According to [58, Th. 3.5], we have

$$\min_{\mathcal{G}^{(n)}} \ \frac{1}{2}\left\|\mathcal{X} - \mathbf{G}_{(2)}^{(n)}\left(\mathbf{G}_{<2>}^{(\neq n)}\right)^{\mathrm{T}}\right\|_F^2 + \frac{\rho}{2}\left\|\mathcal{G}^{(n)} - \mathcal{G}^{(n),l}\right\|_F^2. \tag{7}$$

The least-square problem (7) can be solved as follows:

$$\begin{aligned} \mathcal{G}^{(n),l+1} = \text{Fold}_2\Big(\left(\mathbf{X}_{\langle n\rangle}\mathbf{G}_{\langle 2\rangle}^{(\neq n)} + \rho\mathbf{G}_{(2)}^{(n)}\right) \\ \times\left(\mathbf{G}_{\langle 2\rangle}^{(\neq n),\mathrm{T}}\mathbf{G}_{\langle 2\rangle}^{(\neq n)} + \rho\mathbf{I}\right)^{-1}\Big) \end{aligned} \tag{8}$$

where $\mathbf{I}$ is an identity matrix.

*2) Update $\mathcal{X}$:* The $\mathcal{X}$ subproblem is

$$\begin{aligned} &\min_{\mathcal{X}} \ \frac{1}{2}\left\|\mathcal{X} - \Phi\left(\mathcal{G}^{l+1}\right)\right\|_F^2 + \frac{\alpha}{2}\sum_{k=1}^{2}\|\mathcal{X} \times_k \mathbf{D}_k - \mathcal{Q}_k\|_F^2 \\ &\qquad + \frac{\rho}{2}\left\|\mathcal{X} - \mathcal{X}^l\right\|_F^2 \\ &\text{s.t.} \ \ \mathcal{P}_\Omega(\mathcal{X} - \mathcal{Y}) = 0 \end{aligned} \tag{9}$$

which can be decoupled into $t$ subproblems

$$\begin{aligned} &\min_{\mathcal{X}_t} \ \frac{1}{2}\left\|\mathcal{X}_t - \Phi\left(\mathcal{G}^{l+1}\right)_t\right\|_F^2 + \frac{\alpha}{2}\sum_{k=1}^{2}\|\mathcal{X}_t \times_k \mathbf{D}_k - (\mathcal{Q}_k)_t\|_F^2 \\ &\qquad + \frac{\rho}{2}\left\|\mathcal{X}_t - \mathcal{X}_t^l\right\|_F^2 \\ &\text{s.t.} \ \ \mathcal{P}_\Omega(\mathcal{X}_t - \mathcal{Y}_t) = 0. \end{aligned} \tag{10}$$

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According to Definition 2, (10) is equivalent to

$$\begin{aligned}\min_{\mathbf{X}_{t(3)}^{\mathrm{T}}}\ & \frac{1}{2}\left\|\mathbf{X}_{t(3)}^{\mathrm{T}}-(\Phi(\mathcal{G}^{l+1})_t)_{(3)}^{\mathrm{T}}\right\|_F^2 \\ &+\frac{\alpha}{2}\left\{\left\|(\mathbf{I}_N\otimes\mathbf{D}_1)\mathbf{X}_{t(3)}^{\mathrm{T}}-[(\mathbf{Q}_1)_t]_{(3)}^{\mathrm{T}}\right\|_F^2\right. \\ &\left.+\left\|(\mathbf{D}_2\otimes\mathbf{I}_M)\mathbf{X}_{t(3)}^{\mathrm{T}}-[(\mathbf{Q}_2)_t]_{(3)}^{\mathrm{T}}\right\|_F^2\right\} \\ &+\frac{\rho}{2}\left\|\mathbf{X}_{t(3)}^{\mathrm{T}}-(\mathbf{X}_t^l)_{(3)}^{\mathrm{T}}\right\|_F^2 \\ \text{s.t.}\ & \mathcal{P}_\Omega(\mathbf{X}_{t(3)}^{\mathrm{T}}-\mathbf{Y}_{t(3)}^{\mathrm{T}})=0. \end{aligned} \tag{11}$$

Taking the derivative of the above equation, we have

$$\mathbf{H}\mathbf{X}_{t(3)}^{\mathrm{T}}+\mathbf{X}_{t(3)}^{\mathrm{T}}(1+\rho)\mathbf{I}_B=\mathbf{K} \tag{12}$$

where $\mathbf{H}=\alpha\left[\left(\mathbf{I}_N\otimes\mathbf{D}_1^{\mathrm{T}}\mathbf{D}_1\right)+\left(\mathbf{D}_2^{\mathrm{T}}\mathbf{D}_2\otimes\mathbf{I}_M\right)\right]$ and

$$\begin{aligned}\mathbf{K}=(\Phi(\mathcal{G}^{l+1})_t)_{(3)}^{\mathrm{T}}&+\alpha\left[\left(\mathbf{I}_N\otimes\mathbf{D}_1^{\mathrm{T}}\right)[(\mathbf{Q}_1)_t]_{(3)}^{\mathrm{T}}\right.\\ &\left.+\left(\mathbf{D}_2^{\mathrm{T}}\otimes\mathbf{I}_M\right)[(\mathbf{Q}_2)_t]_{(3)}^{\mathrm{T}}\right]+\rho(\mathbf{X}_t^l)_{(3)}^{\mathrm{T}}.\end{aligned}$$

Since the matrix $\mathbf{H}$ is a block circulant matrix with circulant blocks, $\mathbf{H}$ has the following spectral decomposition, i.e.,

$$\mathbf{H}=\mathbf{F}^{\mathrm{H}}\Sigma\mathbf{F}$$

where $\mathbf{F}$ is the 2-D discrete Fourier transform matrix and $\Sigma$ is the diagonal matrix of eigenvalues of $\mathbf{H}$. $(1+\rho)\mathbf{I}_B$ is equivalent to $\mathbf{U}(1+\rho)\mathbf{I}_B\mathbf{U}^{\mathrm{T}}$, where $\mathbf{U}^{\mathrm{T}}\mathbf{U}=\mathbf{I}$. Based on [64, Th. 1], the solution of (9) is

$$\mathcal{X}_t^{l+1}=\mathcal{P}_\Omega(\mathcal{Y})+\mathcal{P}_{\Omega^c}\left(\mathrm{Fold}_3\left(\left[\mathbf{F}^{\mathrm{H}}((1\oslash\mathbf{T})\odot(\mathbf{FKU}))\mathbf{U}^{\mathrm{T}}\right]^{\mathrm{T}}\right)\right) \tag{13}$$

where

$$\begin{aligned}\mathbf{T}=&(\mathrm{diag}(\Sigma),\mathrm{diag}(\Sigma),\ldots,\mathrm{diag}(\Sigma))\\ &+(\mathrm{diag}(1+\rho)\mathbf{I}_B,\mathrm{diag}(1+\rho)\mathbf{I}_B,\ldots,\mathrm{diag}(1+\rho)\mathbf{I}_B)^{\mathrm{T}}.\end{aligned} \tag{14}$$

The aforementioned pseudocode of the PAM algorithm is summarized in Algorithm 2.

### C. Computational Complexity Analysis

For simplicity, we assume the observed multitemporal RS image $\mathcal{Y}\in\mathbb{R}^{M\times N\times B\times T}$, and the TR rank is $(R,R,R,R)$. At each iteration, the main computational complexity lies in updating the $\mathcal{G}^{(n)}$ $(n=1\text{–}4)$ and $\mathcal{X}$. Specifically, the computational complexity of updating $\mathcal{G}^{(1)}$ is $\mathcal{O}(R^2MNBT+R^4NBT)$. Analogously, the computational complexity of updating $\mathcal{G}^{(2)}$, $\mathcal{G}^{(3)}$, and $\mathcal{G}^{(4)}$ is $\mathcal{O}(R^2MNBT+R^4MBT)$, $\mathcal{O}(R^2MNBT+R^4MNT)$, and $\mathcal{O}(R^2MNBT+R^4MNB)$, respectively. The computational complexity of updating $\mathcal{X}$ is $\mathcal{O}(M^2N^2BT)$. In summary, the total computational complexity at each iteration of Algorithm 2 is $\mathcal{O}(R^4(NBT+MBT+MNT+MNB)+M^2N^2BT)$.

### D. Convergence Analysis

We analyze the convergence of the developed PAM-based solving algorithm. First, the objective function $f(\mathcal{G},\mathcal{X})$ can be reformulated as follows:

$$\begin{aligned}f(\mathcal{G},\mathcal{X})=&\frac{1}{2}\|\mathcal{X}-\Phi(\mathcal{G})\|_F^2+\frac{\alpha}{2}\sum_{k=1}^{2}\|\mathcal{X}\times_k\mathbf{D}_k-\mathcal{Q}_k\|_F^2\\ &+\iota_{\mathbb{S}}(\mathcal{X})\end{aligned} \tag{15}$$

where

$$\iota_{\mathbb{S}}(\mathcal{X}):=\begin{cases}0, & \text{if}\ \ \mathcal{X}\in\mathbb{S}\\ \infty, & \text{otherwise}\end{cases} \tag{16}$$

with $\mathbb{S}:=\{\mathcal{X}:\mathcal{P}_\Omega(\mathcal{X}-\mathcal{F})=0\}$.

Then, we analyze the theoretical convergence guarantee of Algorithm 2 based on Theorem 1.

*Theorem 1:* Assuming the sequence $\{\mathcal{G}^l,\mathcal{X}^l\}_{l\in\mathbb{N}}$ generated by Algorithm 2 is bounded, it converges to a critical point of $f(\mathcal{G},\mathcal{X})$.

*Proof:* Following [63, Th. 6.2], we prove Theorem 1 by verifying that the following four conditions are satisfied.

1) $\|\mathcal{X}-\Phi(\mathcal{G})\|_F^2$ is a $C^1$ function with locally Lipschitz continuous gradient, and $\sum_{k=1}^{2}\|\mathcal{X}\times_k\mathbf{D}_k-\mathcal{Q}_k\|_F^2$ and $\iota_{\mathbb{S}}(\mathcal{X})$ are proper lower semicontinuous functions.
2) $f(\mathcal{G},\mathcal{X})$ has the Kurdyka–Łojasiewicz (K–Ł) property [63] at each $\{\mathcal{G}^l,\mathcal{X}^l\}_{l\in\mathbb{N}}\in\mathrm{dom}(f)$.
3) *Sufficient Decrease Condition:* For $\rho>0$, the sequence $\{\mathcal{G}^l,\mathcal{X}^l\}_{l\in\mathbb{N}}$ satisfies

$$\begin{cases}f\left(\mathcal{G}^{(n),l+1},\mathcal{X}^l\right)+\dfrac{\rho}{2}\left\|\mathcal{G}^{(n),l+1}-\mathcal{G}^{(n),l}\right\|_F^2\\ \qquad\le f\left(\mathcal{G}^{(n),l},\mathcal{X}^l\right),\quad n=1,2,\ldots,N\\ f\left(\mathcal{G}^{l+1},\mathcal{X}^{l+1}\right)+\dfrac{\rho}{2}\left\|\mathcal{X}^{l+1}-\mathcal{X}^l\right\|_F^2\le f\left(\mathcal{G}^{l+1},\mathcal{X}^l\right).\end{cases} \tag{17}$$

4) *Relative Error Condition:* There exist $\mathcal{A}^{l+1}$ and $\mathcal{B}^{l+1}$, such that the sequence $\{\mathcal{G}^l,\mathcal{X}^l\}_{l\in\mathbb{N}}$ satisfies

$$\begin{cases}\left\|\mathcal{A}^{l+1}+\nabla_{\mathcal{G}^{(n)}}f\left(\mathcal{G}^{(n),l+1},\mathcal{X}^l\right)\right\|_F\\ \qquad\le\rho\left\|\mathcal{G}^{(n),l+1}-\mathcal{G}^{(n),l}\right\|_F,\quad n=1,2,\ldots,N\\ \left\|\mathcal{B}^{l+1}+\nabla_{\mathcal{X}}f\left(\mathcal{G}^{l+1},\mathcal{X}^{l+1}\right)\right\|_F\le\rho\left\|\mathcal{X}^{l+1}-\mathcal{X}^l\right\|_F.\end{cases} \tag{18}$$

*For condition (1):* $\|\mathcal{X}-\Phi(\mathcal{G})\|_F^2$ is a Frobenius norm function, which is a $C^1$ function with locally Lipschitz continuous gradient. In addition, $\sum_{k=1}^{2}\|\mathcal{X}\times_k\mathbf{D}_k-\mathcal{Q}_k\|_F^2$ and indicator function $\iota_{\mathbb{S}}(\mathcal{X})$ are proper lower semicontinuous functions. Therefore, condition (1) is satisfied.

*For condition (2):* Following [65], the semialgebraic function has K–Ł property, and the Frobenius norm functions and indicator functions are semialgebraic functions. Since the sum of semialgebraic functions is also semialgebraic, the function $f(\mathcal{G},\mathcal{X})$ satisfies the K–Ł property. Therefore, condition (2) is satisfied.

*For condition (3):* For $\mathcal{G}^{(n)}$ and $\mathcal{X}$ subproblems, $\mathcal{G}^{(n),l+1}$ and $\mathcal{X}^l$ are the corresponding optimal solutions, respectively.

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We have

$$\begin{cases} f\left(\mathcal{G}^{(n),l+1},\mathcal{X}^{l}\right)+\dfrac{\rho}{2}\left\|\mathcal{G}^{(n),l+1}-\mathcal{G}^{(n),l}\right\|_F^2 \\ \qquad\qquad \le f\left(\mathcal{G}^{(n),l},\mathcal{X}^{l}\right)+\dfrac{\rho}{2}\left\|\mathcal{G}^{(n),l}-\mathcal{G}^{(n),l}\right\|_F^2 \\ \qquad\qquad = f\left(\mathcal{G}^{(n),l},\mathcal{X}^{l}\right), \quad n=1,2,\ldots,N \\ f\left(\mathcal{G}^{l+1},\mathcal{X}^{l+1}\right)+\dfrac{\rho}{2}\left\|\mathcal{X}^{l+1}-\mathcal{X}^{l}\right\|_F^2 \\ \qquad\qquad \le f\left(\mathcal{G}^{l+1},\mathcal{X}^{l}\right)+\dfrac{\rho}{2}\left\|\mathcal{X}^{l}-\mathcal{X}^{l}\right\|_F^2 \\ \qquad\qquad = f\left(\mathcal{G}^{l+1},\mathcal{X}^{l}\right). \end{cases} \tag{19}$$

Therefore, condition (3) is satisfied.

*For condition (4):* For each subproblem, we have

$$\begin{cases} 0\in\nabla_{\mathcal{G}^{(n)}} f\left(\mathcal{G}^{(n),l+1},\mathcal{X}^{l}\right)+\rho\left(\mathcal{G}^{(n),l+1}-\mathcal{G}^{(n),l}\right) \\ \qquad n=1,2,\ldots,N \\ 0\in\nabla_{\mathcal{X}} f\left(\mathcal{G}^{l+1},\mathcal{X}^{l+1}\right)+\rho\left(\mathcal{X}^{l+1}-\mathcal{X}^{l}\right). \end{cases} \tag{20}$$

Letting

$$\begin{cases} \mathcal{A}^{l+1}=-\nabla_{\mathcal{G}^{(n)}} f\left(\mathcal{G}^{(n),l+1},\mathcal{X}^{l}\right)-\rho\left(\mathcal{G}^{(n),l+1}-\mathcal{G}^{(n),l}\right) \\ \qquad \in 0, n=1,2,\ldots,N \\ \mathcal{B}^{l+1}=-\nabla_{\mathcal{X}} f\left(\mathcal{G}^{l+1},\mathcal{X}^{l+1}\right)-\rho\left(\mathcal{X}^{l+1}-\mathcal{X}^{l}\right)\in 0 \end{cases} \tag{21}$$

we can find that condition (4) is satisfied.

## IV. Numerical Experiments

This section tests the performance of the proposed TRGFid on simulated and real multitemporal RS images. We select six comparison methods, i.e., Regression, HaLRTC [53], TNN [66], TT [57], TRLRF [60], and TVTR [61]. Regression is a method based on the proposed TRGFid that estimates the image at cloud-contaminated time nodes using the image at cloud-free time nodes. HaLRTC, TNN, and TT are methods based on Tucker decomposition, t-SVD, and TT decomposition with pixel domain fidelity, respectively. TRLRF and TVTR are methods based on TR decomposition with pixel domain fidelity and combined with factors regularization and total variation (TV) regularization, respectively. All parameters of the above methods are manually adjusted to achieve optimal performance following the authors' codes and suggestions. All experiments are performed on a desktop PC running Windows 10 and MATLAB (R2019b) with an Intel Core i7-6700 CPU at 3.40 GHz and 16-GB RAM.

To measure the quality of the reconstructed image $\mathcal{X}\in\mathbb{R}^{M\times N\times B\times T}$ of different methods, we employ the peak signal-to-noise rate (PSNR), the structural similarity (SSIM), and the spectral angle mapping (SAM) as the quantitative evaluation metrics, which are, respectively, defined as follows:

$$\mathrm{PSNR}^t=\frac{1}{B}\sum_{b=1}^{B}10\times\log_{10}\frac{\left(\tilde{\mathbf{X}}_b^t\right)_{\max}^2}{\left\|\mathbf{X}_b^t-\tilde{\mathbf{X}}_b^t\right\|_F^2}$$

$$\mathrm{SSIM}^t=\frac{1}{B}\sum_{b=1}^{B}\frac{\left(2\mu_{\mathbf{X}_b^t}\mu_{\tilde{\mathbf{X}}_b^t}+c_1\right)\left(2\sigma_{\mathbf{X}_b^t\tilde{\mathbf{X}}_b^t}+c_2\right)}{\left(\mu_{\mathbf{X}_b^t}^2+\mu_{\tilde{\mathbf{X}}_b^t}^2+c_1\right)\left(\sigma_{\mathbf{X}_b^t}^2+\sigma_{\tilde{\mathbf{X}}_b^t}^2+c_2\right)}$$

TABLE II

Detailed Information of Multitemporal RS Images for Simulated Experiments

| Dataset | Source | Spatial resolution | Image size | Spectral bands | Time nodes |
|---|---|---|---|---|---|
| *Tokyo* | Sentinel-2 | 10 m | $400\times400$ | 4 | 4 |
| *Morocco* | Sentinel-2 | 10 m | $400\times400$ | 4 | 6 |
| *Brazil* | Sentinel-2 | 10 m | $400\times400$ | 4 | 6 |
| *Bourgogne* | Landsat-8 | 30 m | $400\times400$ | 7 | 6 |

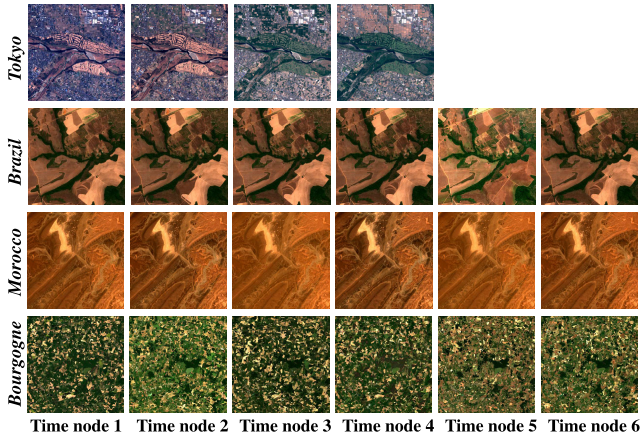



Fig. 2. Multitemporal RS images of four simulated datasets.

and

$$\mathrm{SAM}^t=\frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}\cos^{-1}\left(\frac{\left(\mathbf{X}_{i,j}^t\right)^{\mathrm{T}}\tilde{\mathbf{X}}_{i,j}^t}{\left\|\mathbf{X}_{i,j}^t\right\|_F\left\|\tilde{\mathbf{X}}_{i,j}^t\right\|_F+c_3}\right)$$

where $\tilde{\mathcal{X}}$ is the ground-truth image; $\left(\tilde{\mathbf{X}}_b^t\right)_{\max}$ is the maximum pixel value of $\tilde{\mathbf{X}}_b^t$; $\sigma_{\mathbf{X}_b^t}$ and $\sigma_{\tilde{\mathbf{X}}_b^t}$ are the standard deviation of $\mathbf{X}_b^t$ and $\tilde{\mathbf{X}}_b^t$, respectively; $\sigma_{\mathbf{X}_b^t\tilde{\mathbf{X}}_b^t}$ is the covariance between $\mathbf{X}_b^t$ and $\tilde{\mathbf{X}}_b^t$; $\mu_{\mathbf{X}_b^t}$ and $\mu_{\tilde{\mathbf{X}}_b^t}$ are the average values of $\mathbf{X}_b^t$ and $\tilde{\mathbf{X}}_b^t$, respectively; $\mathbf{X}_{i,j}^t$ and $\tilde{\mathbf{X}}_{i,j}^t$ are the column obtained by fixing indexes $i$, $j$, and $t$ of $\mathcal{X}$ and $\tilde{\mathcal{X}}$, respectively; and $c_1$, $c_2$, and $c_3$ are constants.

### A. Simulated Experiments

*1) Datasets:* We use multitemporal RS images to test the performance of the TRGFid method on four datasets. Three datasets named *Tokyo*,[2] *Brazil*,[2] and *Morocco*[2] are taken over Tokyo, Brazil, and Morocco by Sentinel-2, respectively. One dataset named *Bourgogne*[3] is taken over Bourgogne by Landsat-8. The datasets for simulated experiments are detailed in Table II and Fig. 2. We simulate clouds of different sizes and shapes and add them to the multitemporal RS images, The remaining images are employed as reference images. For fairness, we use the same number of reference images in both the proposed method and the comparison methods. For the *Tokyo* dataset, images at two time nodes are selected to add clouds. For the *Brazil*, *Morocco*, and *Bourgogne* datasets, images at three time nodes are selected to add clouds. All bands of the simulated dataset at the same time node are

[2]https://earthexplorer.usgs.gov
[3]https://theia.cnes.fr/atdistrib/rocket/#/home

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covered by clouds at the same location, while the simulated dataset at different time nodes is covered by clouds at different locations.

*2) Quantitative Comparison:* To quantitatively test the performance of the proposed TRGFid, we show the metric values of different methods in Table III, where the best and second-best results are highlighted in bold and underlined, respectively. As observed, the proposed TRGFid consistently achieves the best results in all simulated experiments and obtains an average PSNR gain of about 3 dB over the second-best method. Although the running time of the proposed TRGFid is not the shortest, it is acceptable considering that the PSNR, SSIM, and SAM values of the proposed method are significantly better than its comparison methods. It should be noted that the proposed TRGFid shows significant improvements when large clouds contaminate multitemporal RS images. This is because the gradient domain fidelity can fully utilize the gradient information of multitemporal RS images at cloud-free time nodes.

*3) Qualitative Comparison:* To qualitatively test the performance of the proposed TRGFid, we show the visual comparison of the results in Figs. 3–6, where the images are enlarged for a clear comparison in the red box. As observed, the proposed TRGFid obtains the best visual results with rich details and textures. Specifically, the proposed TRGFid can reconstruct the cloudy region, which is closest to ground-truth images among the compared methods. Comparatively, the reconstructed results obtained by the other methods remain blurriness, and the reconstructed edges are severely damaged. This is because the guided gradient estimated by the cloud-free region can capture clear and accurate edge features of the cloudy region, and the gradient domain fidelity with respect to the guided gradient can effectively preserve the fine edges and textures compared with the original pixel domain fidelity.

Fig. 7 plots scatter diagrams between the original and reconstructed pixels for all simulated datasets at one time node. As observed, the pixel values reconstructed by our method approximate the original pixel values and show a compact distribution, indicating the superiority of our method. Comparatively, the pixel values reconstructed by TT, TRLRF, and TVTR obviously deviate from the original pixel values. For HaLRTC and TNN, some reconstructed pixel values deviate from the original pixel values, which indicates inaccurate reconstruction in the cloudy regions. For Regression, the reconstructed pixel values show a distribution similar to the TRGFid method, which indicates the effectiveness of using the Regression method to estimate the guided gradient. This further confirms the effectiveness of the TRGFid method and is consistent with the quantitative performance in Table III.

## B. Real Experiments

We select two real datasets to test the performance of the TRGFid method. One dataset named *Bourgogne*[3] is taken over Bourgogne by Landsat-8, and the corresponding cloud masks are also available from the website.[3] Another dataset named *Beijing*[4] is taken over Beijing by SPOT-5, and the corresponding cloud masks are available from the website.[5] The details are given in Table IV and Fig. 8. For the real experiments, the images are contaminated by clouds at the first two time nodes, and the remaining images are used as reference images to estimate the guided gradient.

Figs. 9 and 10 show the cloud removal results, which are enlarged for a clear comparison in the red box. As observed, the results obtained by TT, TRLRF, and TVTR lose image details and textures, while HaLRTC reconstructs the overall structure but loses some details. Although TNN and Regression yield promising results, some fine edges and textures are still blurred in the reconstructed regions. Comparatively, the proposed TRGFid has significant advantages over other compared methods, especially in preserving edges and textures. This is because the gradient domain fidelity can fully use the gradient information of multitemporal RS images at cloud-free time nodes to preserve fine edges and textures.

## C. Discussion

*1) Gradient Domain Fidelity Versus Original Pixel Domain Fidelity:* This section tests the influence of the gradient domain fidelity in the proposed TRGFid, the reconstructed results based on the *Brazil* dataset, and the corresponding PSNR values, as shown in Fig. 11. As observed, the gradient domain fidelity has a powerful ability to preserve fine edges and textures. The results are blurred and unclear when TR decomposition only with original pixel domain fidelity for thick cloud removal. Benefitting from the gradient domain fidelity, our method (the TR decomposition with original pixel domain and gradient domain fidelity) produces the most visually pleasing results with fine textures and edges. In summary, when taking the gradient domain and the original pixel domain fidelity into consideration, the results are better than that isolatedly using one domain fidelity. The abovementioned observation powerfully illustrates the effectiveness of the gradient domain fidelity as a complement.

*2) Gradient Domain Fidelity Versus TV Regularization:* This section compares the difference between gradient domain fidelity and traditional TV regularization.

1) The traditional TV regularization $\sum_{k=1}^{2} \|\mathcal{X} \times_k \mathbf{D}_k\|_1$ utilizes the spatial piecewise smoothness prior (i.e., sparsity prior of gradient) of the underlying RS images, which can enhance the local continuity of images but cannot preserve details well.
2) The gradient domain fidelity $\sum_{k=1}^{2} \|\mathcal{X} \times_k \mathbf{D}_k - \mathcal{Q}_k\|_F^2$ utilizes the gradient information of the image at cloud-free time node, which can guide our method to preserve fine edges and textures.

To verify the superiority of the gradient domain fidelity numerically, we conduct an experiment on the *Brazil* dataset and report the results in Fig. 11. As observed, the reconstructed result of TR decomposition with original pixel domain and gradient domain fidelity is much better at preserving fine edges and textures.

[4]https://take5.theia.cnes.fr/atdistrib/take5/client/#/home

[5]http://www.cesbio.ups-tlse.fr/multitemp/?p=6203

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TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS FOR SIMULATED DATASETS

| Dataset | Time node | Index | Observed | Regression | HaLRTC | TNN | TT | TRLRF | TVTR | TRGFid |
|---|---|---|---|---|---|---|---|---|---|---|
| *Tokoy* | Time node 1 | PSNR | 12.489 | 36.750 | 36.601 | 37.658 | 36.335 | 37.098 | 37.068 | **39.658** |
| | | SSIM | 0.907 | 0.979 | 0.967 | 0.971 | 0.960 | 0.963 | 0.962 | **0.980** |
| | | SAM | 0.898 | 1.982 | 0.488 | 0.506 | 0.530 | 0.488 | 0.538 | **0.350** |
| | Time node 2 | PSNR | 15.813 | 39.367 | 41.590 | 41.475 | 39.995 | 41.185 | 41.506 | **43.589** |
| | | SSIM | 0.952 | 0.990 | 0.984 | 0.985 | 0.981 | 0.982 | 0.982 | **0.990** |
| | | SAM | 0.992 | 1.738 | 0.310 | 0.305 | 0.380 | 0.334 | 0.312 | **0.213** |
| | | Time (s) | — | **0.147** | 80.494 | 86.511 | 47.484 | 22.003 | 36.909 | 87.030 |
| *Brazil* | Time node 1 | PSNR | 9.960 | 42.126 | 39.880 | 40.078 | 34.505 | 38.370 | 38.903 | **45.166** |
| | | SSIM | 0.830 | 0.993 | 0.972 | 0.984 | 0.935 | 0.953 | 0.956 | **0.993** |
| | | SAM | 6.141 | 0.740 | 0.584 | 0.503 | 1.098 | 0.628 | 0.677 | **0.274** |
| | Time node 2 | PSNR | 8.077 | 39.696 | 36.704 | 40.283 | 33.359 | 37.454 | 37.520 | **43.664** |
| | | SSIM | 0.765 | 0.992 | 0.955 | 0.982 | 0.923 | 0.944 | 0.943 | **0.994** |
| | | SAM | 9.345 | 0.792 | 1.083 | 0.635 | 1.582 | 0.834 | 0.882 | **0.377** |
| | Time node 3 | PSNR | 7.185 | 39.296 | 34.622 | 34.180 | 31.367 | 34.867 | 36.237 | **43.095** |
| | | SSIM | 0.700 | 0.987 | 0.932 | 0.961 | 0.885 | 0.916 | 0.922 | **0.988** |
| | | SAM | 11.710 | 0.784 | 1.415 | 1.293 | 2.038 | 1.223 | 1.202 | **0.430** |
| | | Time (s) | — | **0.198** | 116.162 | 104.510 | 45.334 | 138.109 | 234.909 | 256.281 |
| *Morocco* | Time node 1 | PSNR | 12.394 | 46.175 | 48.771 | 48.074 | 41.122 | 44.082 | 44.683 | **52.740** |
| | | SSIM | 0.892 | 0.998 | 0.995 | 0.996 | 0.976 | 0.984 | 0.986 | **0.998** |
| | | SAM | 2.571 | 0.311 | 0.115 | 0.129 | 0.184 | 0.126 | 0.140 | **0.087** |
| | Time node 2 | PSNR | 11.813 | 42.628 | 44.872 | 46.690 | 38.323 | 42.253 | 42.506 | **48.417** |
| | | SSIM | 0.890 | 0.995 | 0.990 | 0.992 | 0.963 | 0.976 | 0.979 | **0.995** |
| | | SAM | 2.868 | 0.371 | 0.171 | 0.124 | 0.186 | 0.158 | 0.190 | **0.108** |
| | Time node 3 | PSNR | 5.934 | 38.322 | **41.675** | 41.296 | 37.473 | 40.340 | 40.091 | 40.769 |
| | | SSIM | 0.677 | 0.983 | 0.972 | 0.977 | 0.955 | 0.956 | 0.959 | **0.984** |
| | | SAM | 10.433 | 0.406 | 0.544 | 0.574 | 0.425 | 0.428 | 0.583 | **0.367** |
| | | Time (s) | — | **0.169** | 118.563 | 80.548 | 175.370 | 85.336 | 90.260 | 304.307 |
| *Bourgogne* | Time node 1 | PSNR | 11.081 | 32.921 | 36.979 | 35.581 | 36.085 | 36.093 | 36.420 | **37.386** |
| | | SSIM | 0.869 | 0.950 | 0.954 | 0.949 | 0.946 | 0.945 | 0.948 | **0.964** |
| | | SAM | 4.777 | 3.090 | 1.207 | 1.281 | 1.417 | 1.318 | 1.263 | **1.135** |
| | Time node 2 | PSNR | 14.108 | 37.084 | 37.264 | 36.560 | 35.883 | 36.334 | 36.543 | **38.264** |
| | | SSIM | 0.933 | 0.981 | 0.973 | 0.973 | 0.967 | 0.967 | 0.968 | **0.983** |
| | | SAM | 2.555 | 2.747 | 0.600 | 0.663 | 0.684 | 0.669 | 0.649 | **0.487** |
| | Time node 3 | PSNR | 11.807 | 37.145 | 37.763 | 36.743 | 35.696 | 36.379 | 36.254 | **38.755** |
| | | SSIM | 0.885 | 0.971 | 0.968 | 0.965 | 0.952 | 0.955 | 0.954 | **0.975** |
| | | SAM | 3.916 | 3.050 | 0.997 | 1.100 | 1.273 | 1.214 | 1.288 | **0.983** |
| | | Time (s) | — | **0.293** | 236.599 | 159.667 | 73.821 | 11.775 | 25.889 | 255.249 |
| | Average | PSNR | 10.969 | 39.276 | 39.702 | 39.874 | 36.382 | 38.587 | 38.885 | **42.866** |
| | | SSIM | 0.845 | 0.984 | 0.969 | 0.976 | 0.949 | 0.958 | 0.960 | **0.986** |
| | | SAM | 5.110 | 1.443 | 0.683 | 0.647 | 0.891 | 0.675 | 0.702 | **0.437** |
| | | Time (s) | — | **0.202** | 137.955 | 107.809 | 85.502 | 64.306 | 96.992 | 255.717 |

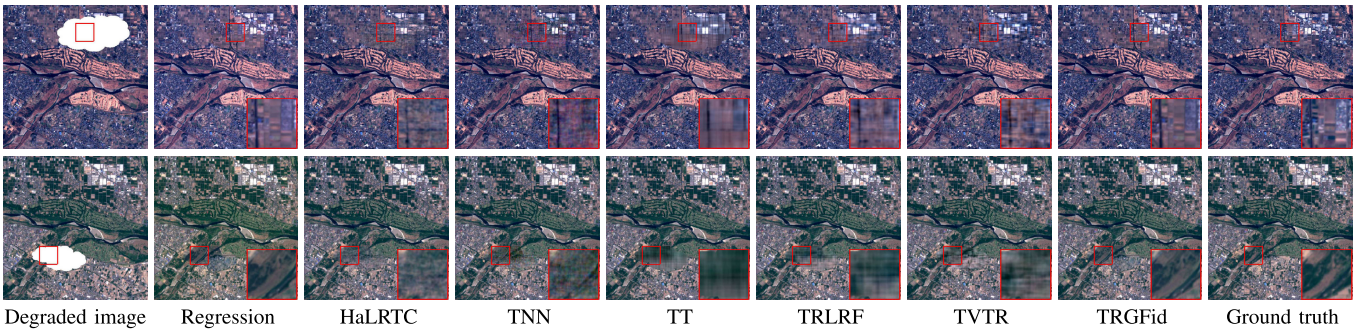



Fig. 3. Cloud removal results of different methods on the *Tokyo* dataset. From top to bottom: visual results (R: B3, G: B2, and B: B1) at time nodes 1 and 2, respectively.

*3) Influences of Guided Gradient:* We discuss the influences of guided gradient estimation strategies. Specifically, we compare the proposed strategy with two alternative strategies: 1) nearest, which uses the cloud-free image that is the nearest in time to the cloud-contaminated image and 2) mean, which uses the mean of the cloud-free images. The quantitative results of different guided gradient estimation strategies are shown in Table V, and we highlight the best values in bold. As observed,

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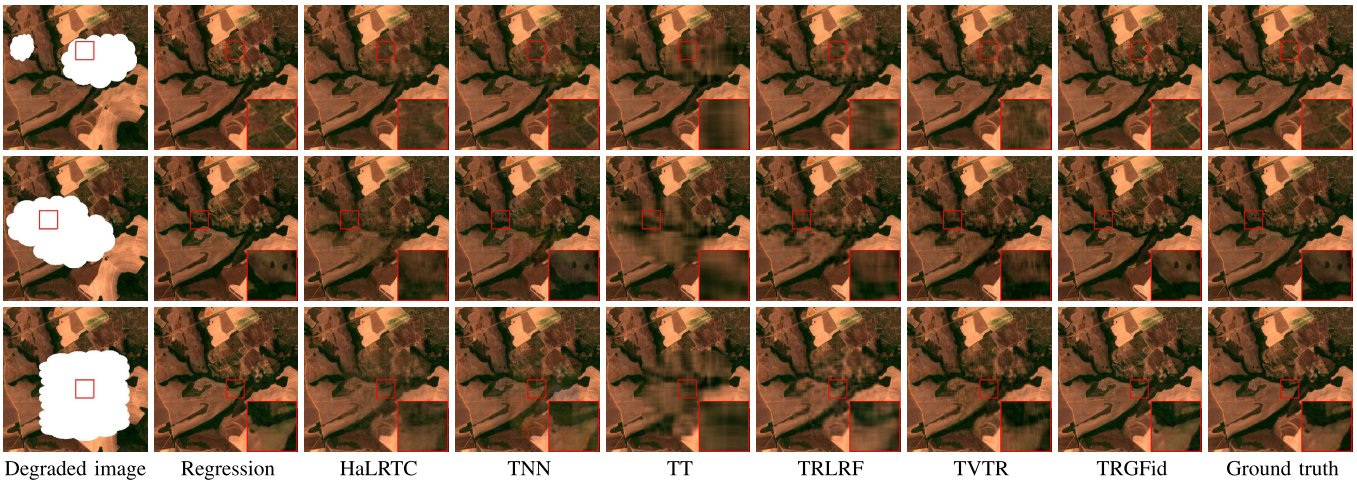



Fig. 4. Cloud removal results of different methods on the *Brazil* dataset. From top to bottom: visual results (R: B3, G: B2, and B: B1) at time nodes 1–3, respectively.

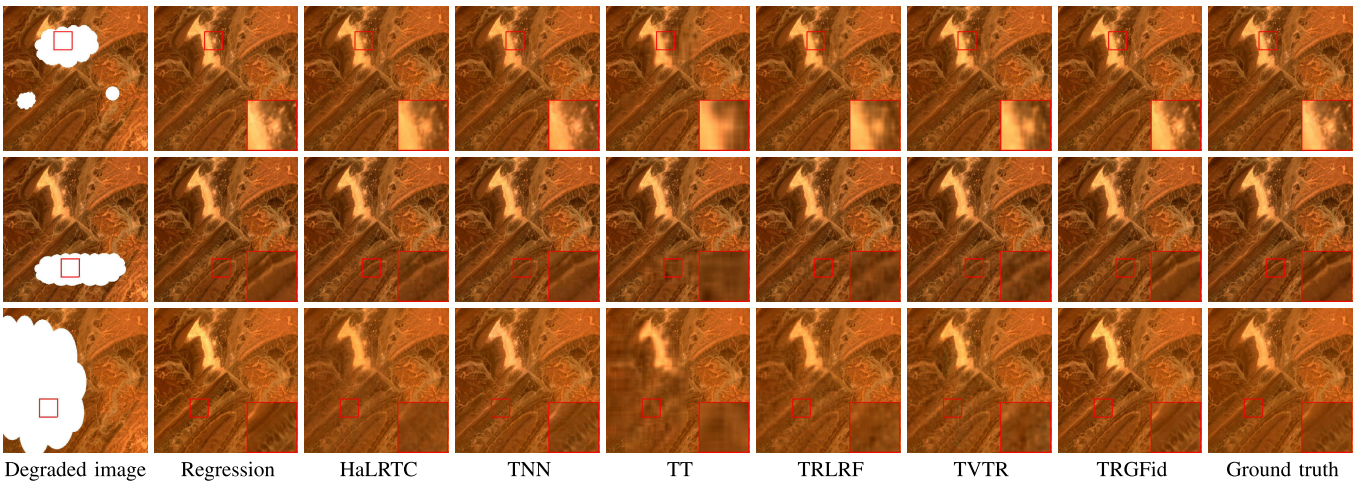



Fig. 5. Cloud removal results of different methods on the *Morocco* dataset. From top to bottom: visual results (R: B3, G: B2, and B: B1) at time nodes 1–3, respectively.

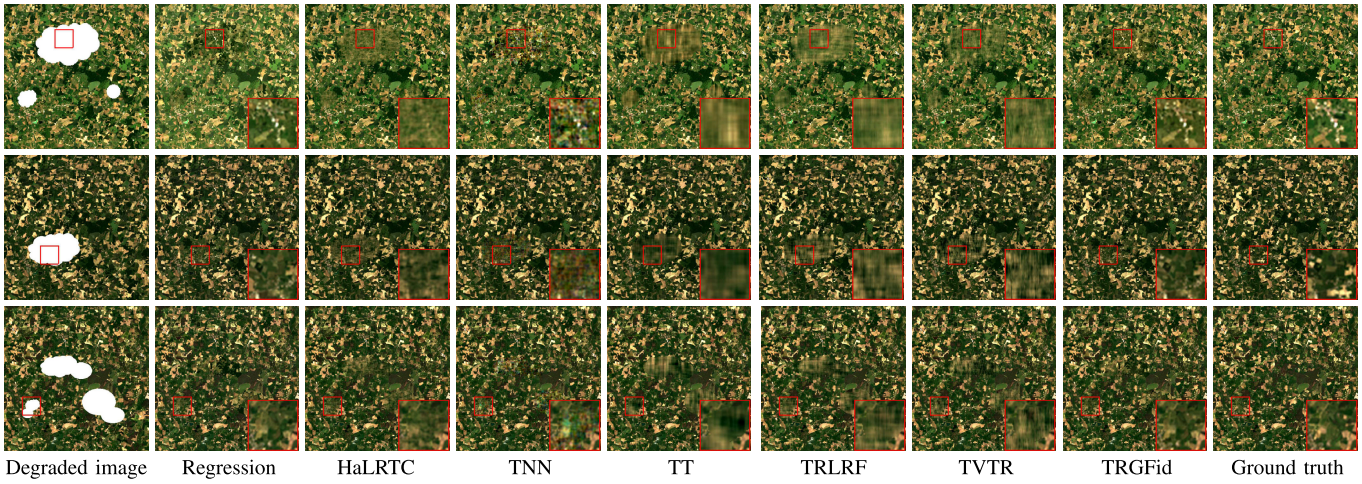



Fig. 6. Cloud removal results of different methods on the *Bourgogne* dataset. From top to bottom: visual results (R: B4, G: B3, and B: B2) at time nodes 1–3, respectively.

the estimation strategies in the proposed TRGFid can promote the average PSNR by almost 1 dB. This implies that our Regression estimation strategy is more effective in estimating the guided gradient.

*4) Influences of Reference Images:* We analyze the influences of the number of reference images. Specifically, we evaluated the performance of the proposed method with one, two, and three reference images, respectively, and present the

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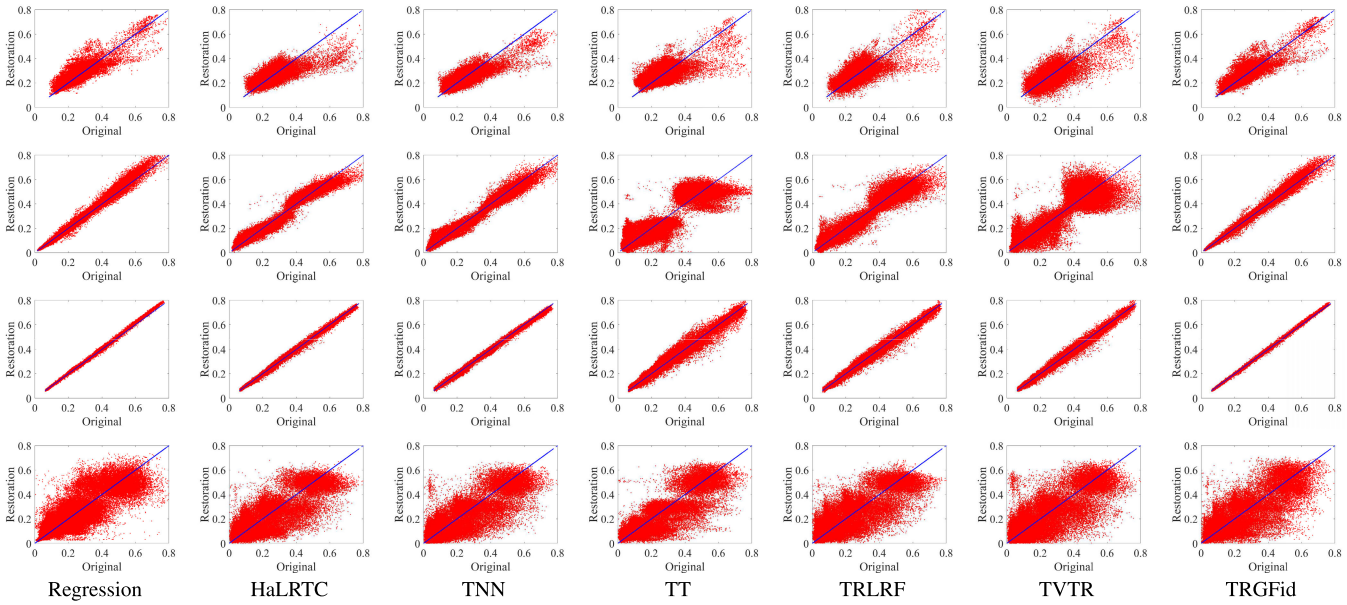



Fig. 7. Scatter diagrams between the original and reconstructed pixels of the cloudy regions. From top to bottom: the *Tokyo* dataset at the first time node, the *Brazil* dataset at the first time node, the *Morocco* dataset at the first time node, and the *Bourgogne* dataset at the second time node, respectively.

TABLE IV

DETAILED INFORMATION OF MULTITEMPORAL RS IMAGES FOR REAL EXPERIMENTS

| Dataset | Source | Spatial resolution | Image size | Spectral bands | Time nodes |
|---|---|---|---|---|---|
| *Bourgogne* | Landsat-8 | 30 m | 600 × 600 | 7 | 4 |
| *Beijing* | SPOT-5 | 10 m | 1000 × 1000 | 4 | 4 |

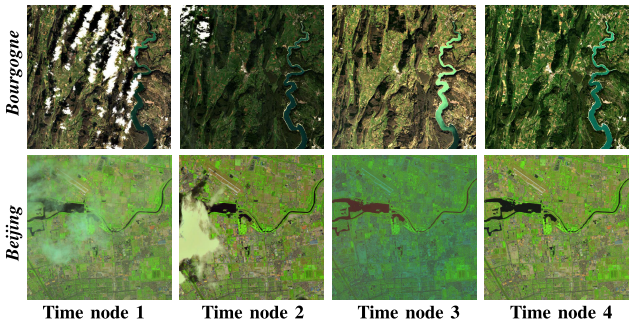



Fig. 8. Multitemporal RS images of two real datasets.

quantitative results based on the *Brazil* dataset in Table VI. As observed, increasing the number of reference images can further improve the performance of the proposed method by obtaining a more accurate guided gradient. However, this comes at the cost of a slight increase in running time. Therefore, finding a balance between performance and time consumption is crucial when choosing the number of reference images. In all experiments, the number of reference images is equal to that of the cloud-contaminated images.

*5) Parameter Analysis:* This section analyzes the influences of the parameters $\mathbf{r}$ and $\alpha$ in the proposed TRGFid. The TR rank $\mathbf{r}$ is used to adjust the global low rankness of multitemporal RS images, and the regularization parameter $\alpha$ determines the weight of the gradient domain fidelity.

TABLE V

QUANTITATIVE RESULTS OF THE PROPOSED METHOD WITH DIFFERENT GUIDED GRADIENT ESTIMATION STRATEGIES

| Time node | Index | Observed | Nearest | Mean | Ours |
|---|---|---|---|---|---|
| Time node 1 | PSNR | 12.394 | 48.751 | 51.635 | **52.740** |
| | SSIM | 0.892 | 0.995 | 0.998 | **0.998** |
| | SAM | 2.571 | 0.099 | **0.086** | 0.087 |
| Time node 2 | PSNR | 11.813 | 45.170 | **48.544** | 48.417 |
| | SSIM | 0.890 | 0.991 | **0.995** | 0.995 |
| | SAM | 2.868 | 0.132 | **0.106** | 0.108 |
| Time node 3 | PSNR | 5.934 | **44.155** | 38.840 | 40.769 |
| | SSIM | 0.677 | **0.985** | 0.983 | 0.984 |
| | SAM | 10.433 | 0.474 | **0.358** | 0.367 |
| Average | PSNR | 10.047 | 46.025 | 46.336 | **47.306** |
| | SSIM | 0.820 | 0.990 | 0.992 | **0.992** |
| | SAM | 5.291 | 0.235 | **0.183** | 0.187 |

TABLE VI

QUANTITATIVE RESULTS OF THE PROPOSED METHOD WITH DIFFERENT NUMBERS OF REFERENCE IMAGES

| Time node | Index | Observed | Number of reference images | | |
|---|---|---|---|---|---|
| | | | 1 | 2 | 3 |
| Time node 1 | PSNR | 9.960 | 40.060 | 43.139 | **45.444** |
| | SSIM | 0.830 | 0.980 | 0.990 | **0.994** |
| | SAM | 6.141 | 0.541 | 0.343 | **0.264** |
| Time node 2 | PSNR | 8.077 | 37.028 | 40.987 | **44.173** |
| | SSIM | 0.765 | 0.978 | 0.990 | **0.994** |
| | SAM | 9.345 | 0.691 | 0.443 | **0.342** |
| Time node 3 | PSNR | 7.185 | 29.909 | 40.212 | **42.546** |
| | SSIM | 0.700 | 0.941 | 0.982 | **0.988** |
| | SAM | 11.710 | 0.964 | 0.498 | **0.437** |
| Average | PSNR | 8.407 | 35.666 | 41.446 | **44.054** |
| | SSIM | 0.765 | 0.975 | 0.986 | **0.992** |
| | SAM | 9.065 | 0.620 | 0.419 | **0.348** |
| | Time (s) | — | **218.218** | 242.506 | 261.832 |

We first analyze the influence of the TR rank $\mathbf{r}$. For simplicity, we set the two spatial dimensions of $\mathbf{r}$ to the same value, i.e., $R_1 = R_2$, and the spectral and temporal

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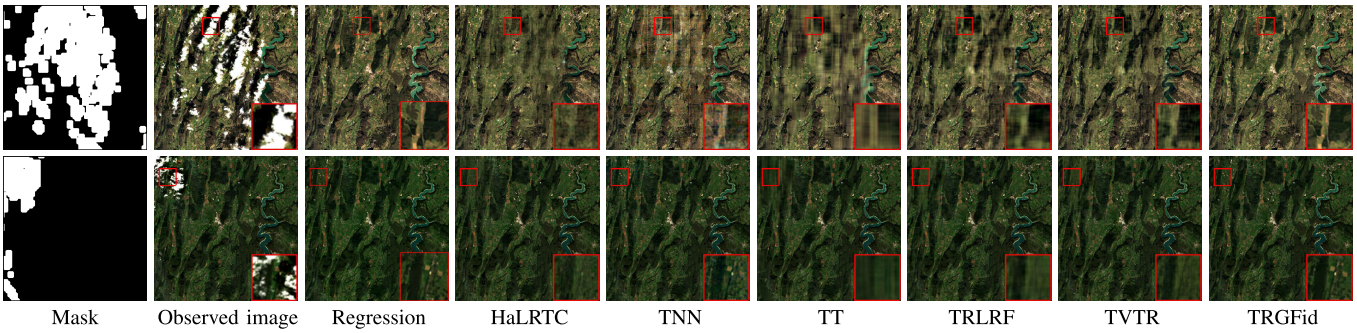



Fig. 9. Cloud removal results of different methods on the real *Bourgogne* dataset. From top to bottom: visual results (R: B4, G: B3, and B: B2) at time nodes 1 and 2, respectively.

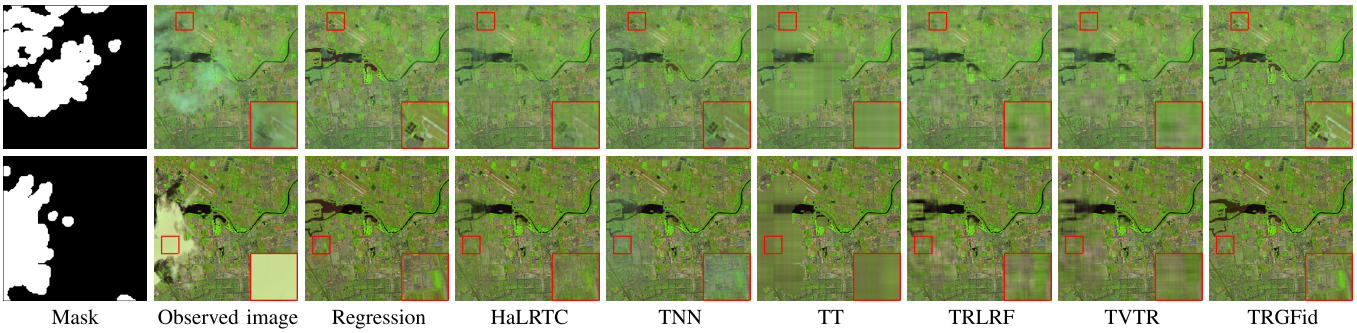



Fig. 10. Cloud removal results of different methods on the real *Beijing* dataset. From top to bottom: visual results (R: B4, G: B3, and B: B2) at time nodes 1 and 2, respectively.

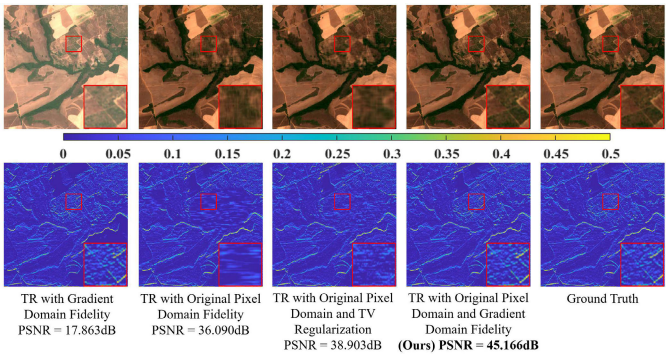



Fig. 11. Reconstructed results by different methods on the *Brazil* dataset. The first and second rows are the reconstructed results and the corresponding gradient along the spatial horizontal direction of the images, respectively.

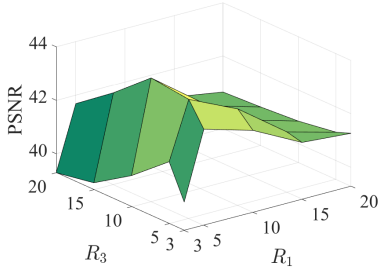



Fig. 12. Influence of the parameter **r** on the *Brazil* dataset.

dimensions of **r** to the same value, i.e., $R_3 = R_4$. Fig. 12 shows the PSNR values on the *Brazil* dataset when we select $R_1$ and $R_2$ from the candidate set {3, 5, 10, 15, 20}. As observed, the proposed TRGFid method exhibits stable and superior performance. Taking into account both experimental performance and computational complexity, the parameter **r** is set to be (5, 5, 10, 10) in all experiments.

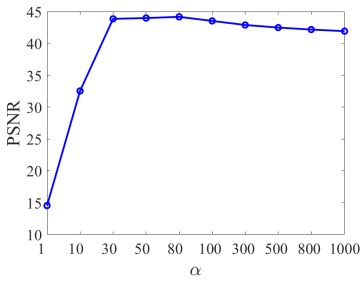



Fig. 13. Influence of the parameter $\alpha$ on the *Brazil* dataset.

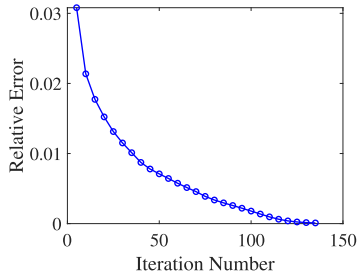



Fig. 14. Relative error versus with respect to the iteration number on the *Brazil* dataset.

Subsequently, we analyze the influence of the parameter $\alpha$ in the TRGFid method. Fig. 13 shows the influence of $\alpha$ on PSNR values when we select $\alpha$ from the candidate set {1, 10, 30, 50, 80, 100, 300, 500, 800, 1000}. For the *Brazil*

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dataset, the PSNR value rises rapidly as $\alpha$ ranges from 1 to 30; then, the value rises slowly before gradually tending to a steady decline, and the best performance of cloud removal is achieved at $\alpha = 80$. In the TRGFid method, the parameter $\alpha$ is set to be 80 in all experiments.

*6) Convergence Analysis:* We analyze the convergence of the developed PAM-based solving algorithm numerically. Fig. 14 shows the relative error of the reconstructed image at each iteration and its previous iteration on the *Brazil* dataset. It is observed that as the iteration number increases, the relative error continuously decreases and eventually converges to zero. This verifies the convergence of the developed algorithm numerically.

## V. Conclusion

In this article, we have suggested a gradient domain fidelity with respect to the guided gradient to guide the reconstruction of the cloudy region. It complements the original pixel domain fidelity by faithfully preserving edges and textures. Armed with the gradient domain fidelity, we proposed a low-rank TR decomposition model for thick cloud removal in multitemporal RS images. The gradient domain fidelity and the low-rank TR regularization are organically integrated to deliver promising results with fine edges and textures. Moreover, we developed an efficient PAM-based algorithm to solve the proposed nonconvex model. Extensive numerical experiments in simulated and real cloud-contamination scenarios showed that the proposed TRGFid outperforms its competitors, especially in edge and texture preservation.

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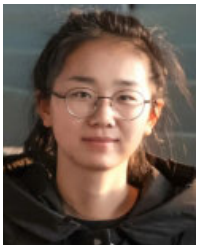
Dr. Huang is an Editor of *The Scientific World Journal*, *Advances in Numerical Analysis*, the *Journal of Applied Mathematics*, the *Journal of Pure and Applied Mathematics: Advances in Applied Mathematics*, and the *Journal of Electronic Science and Technology* (China).

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